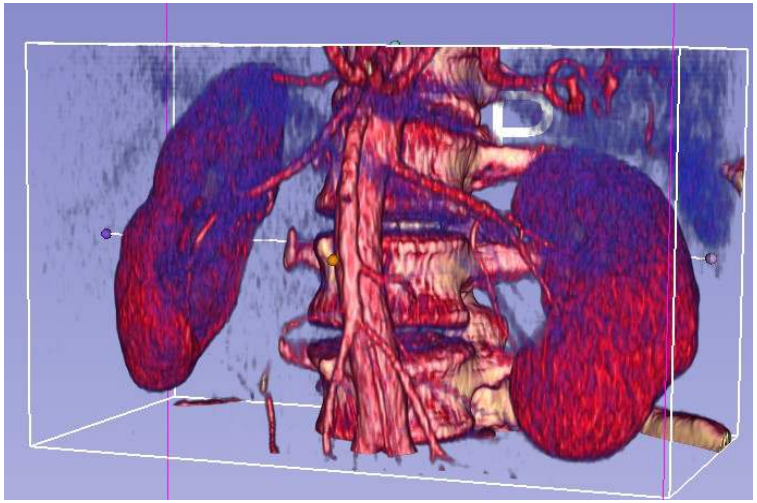
3D Visualization of DICOM Data

Sonia Pujol, Ph.D.

Assistant Professor of Radiology
Director, 3D Slicer Training & Education
Brigham and Women's Hospital
Harvard Medical School

spujol@bwh.harvard.edu

Goal of the Tutorial



The tutorial guides you step-by-step through the process of loading and visualizing a DICOM CT dataset with 3D Slicer.

© Sonia Pujol, Ph.D. All Rights Reserved

Tutorial Materials

- Software:

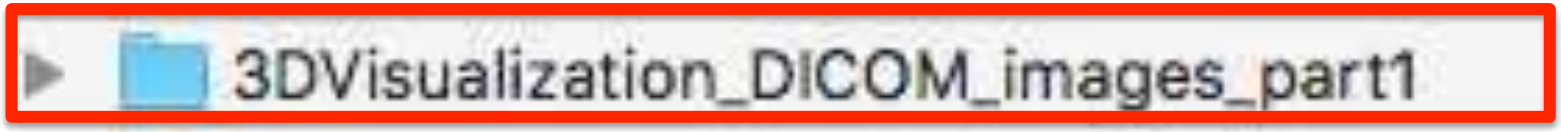
3D Slicer version 4.8.1

- Dataset:

3D Visualization DICOM images part 1

© Sonia Pujol, Ph.D. All Rights Reserved

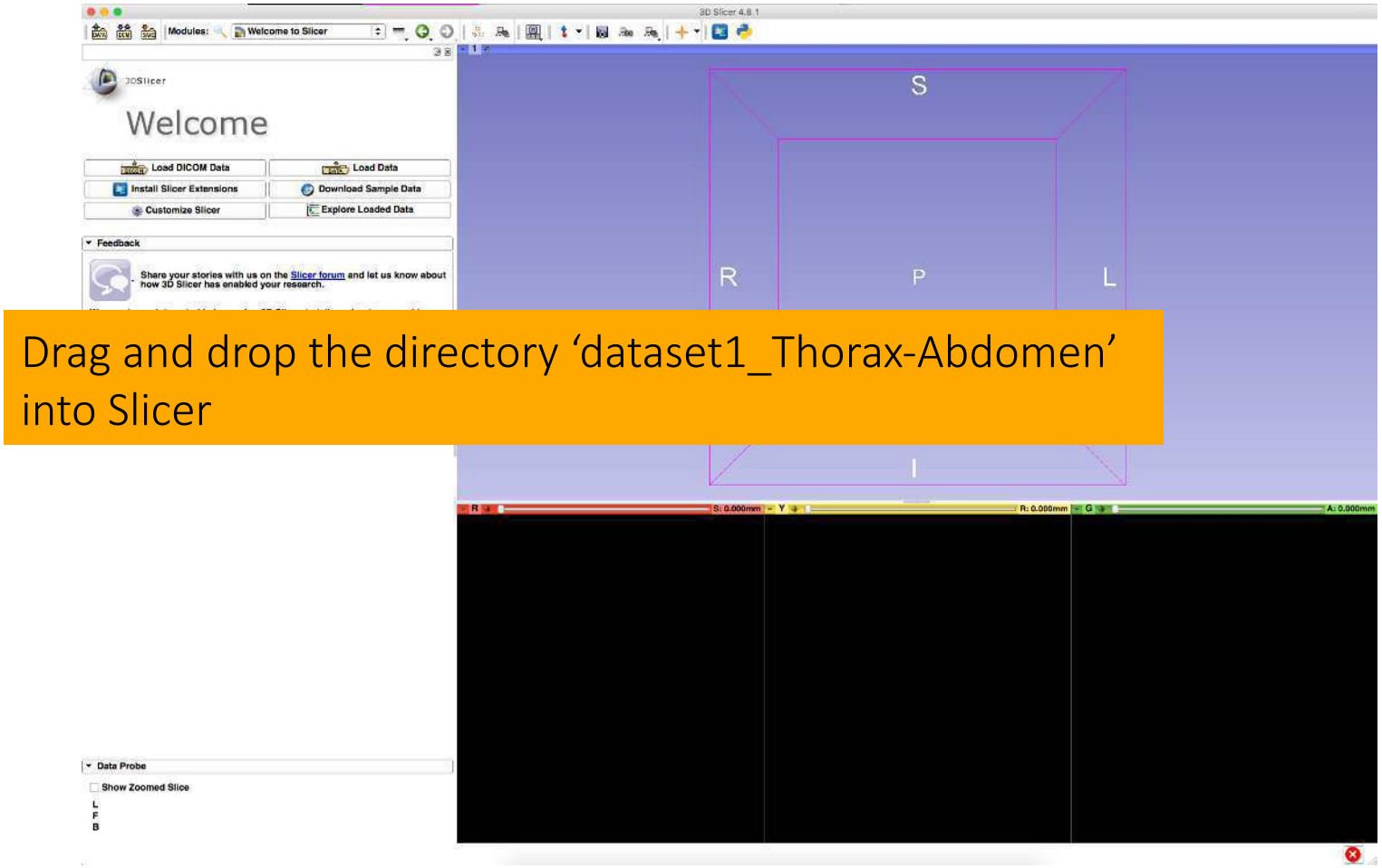
Tutorial Dataset



Download and unzip the file
3DVisualization_DICOM_images_part1.zip

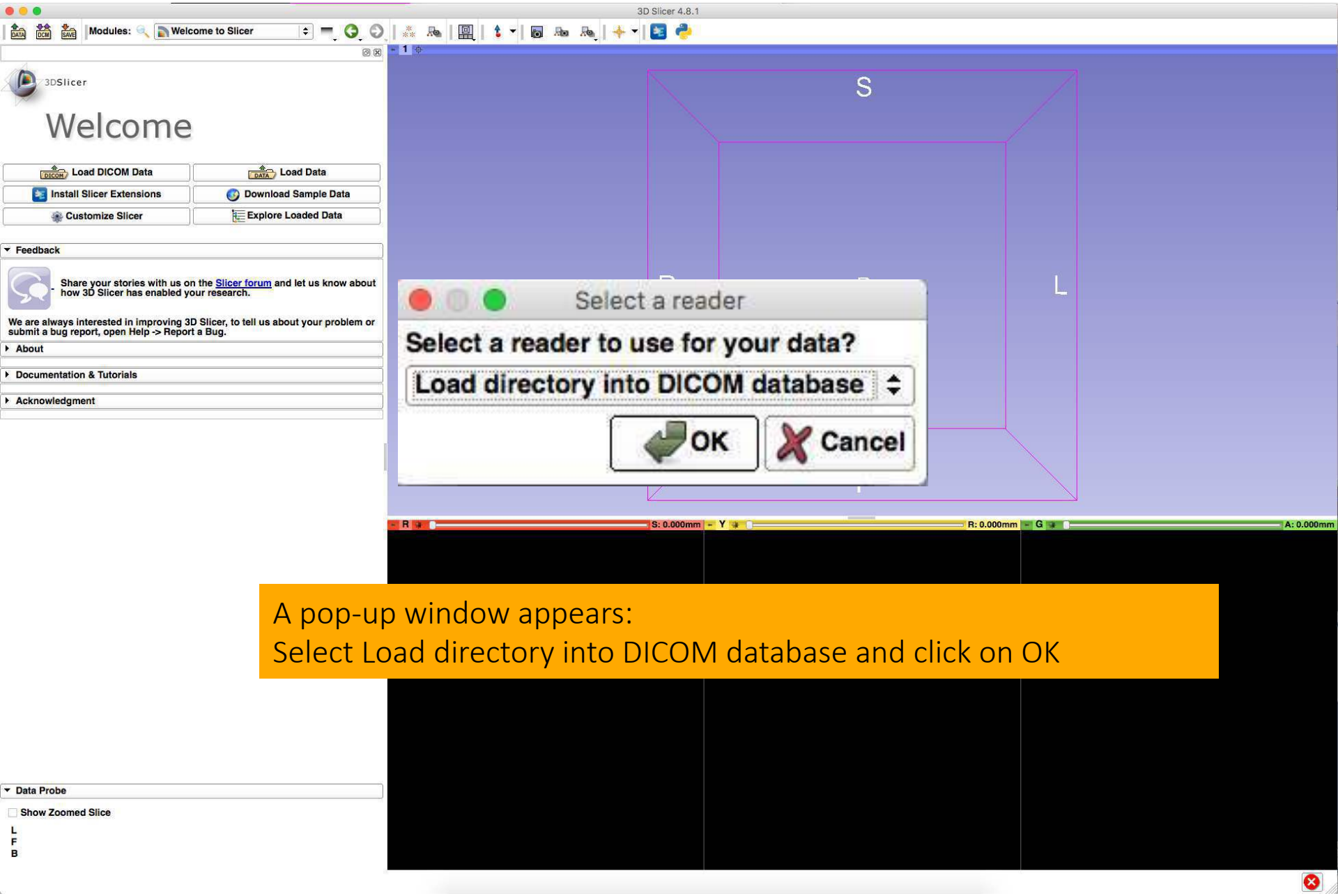
© Sonia Pujol, Ph.D. All Rights Reserved

Loading a DICOM volume



Drag and drop the directory ‘dataset1_Thorax-Abdomen’ into Slicer

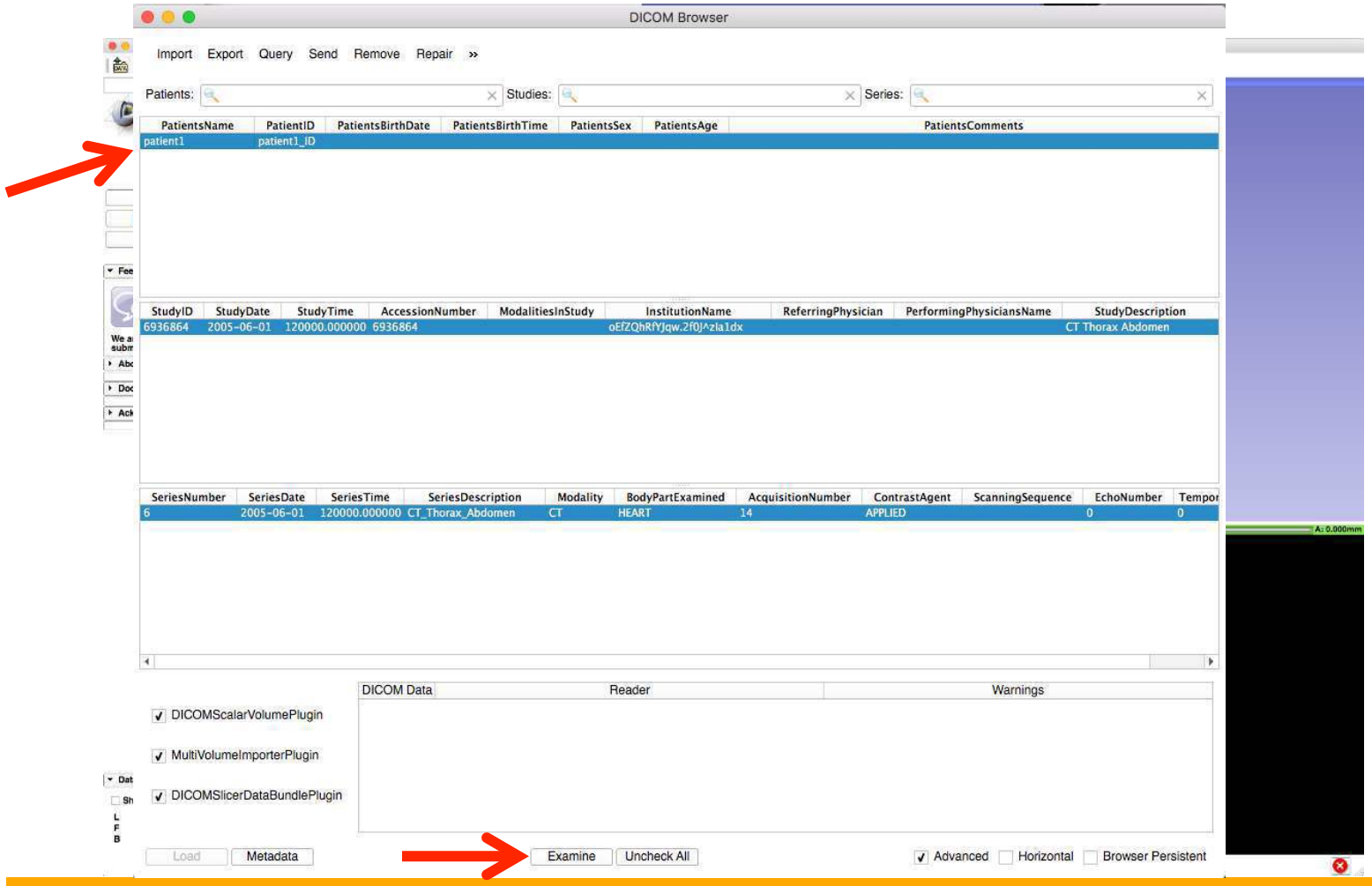
© Sonia Pujol, Ph.D. All Rights Reserved



A pop-up window appears:
Select Load directory into DICOM database and click on OK

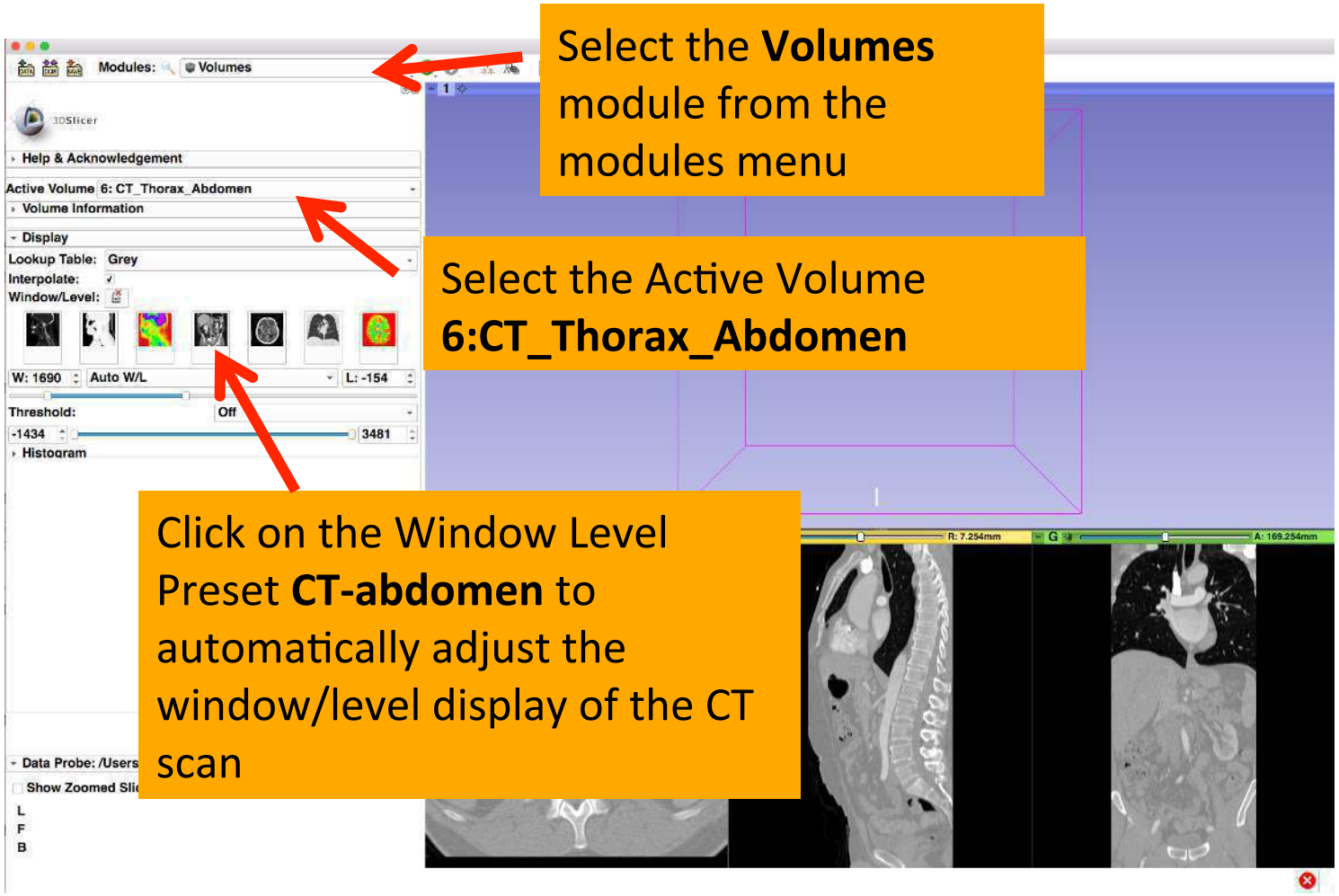
© Sonia Pujol, Ph.D. All Rights Reserved

Loading a DICOM volume



The **patient1** DICOM dataset appears in the DICOM browser.
Select **patient1**, click on **Examine** and then click **Load** to import the data into Slicer

Loading a DICOM volume



Select the **Volumes** module from the modules menu

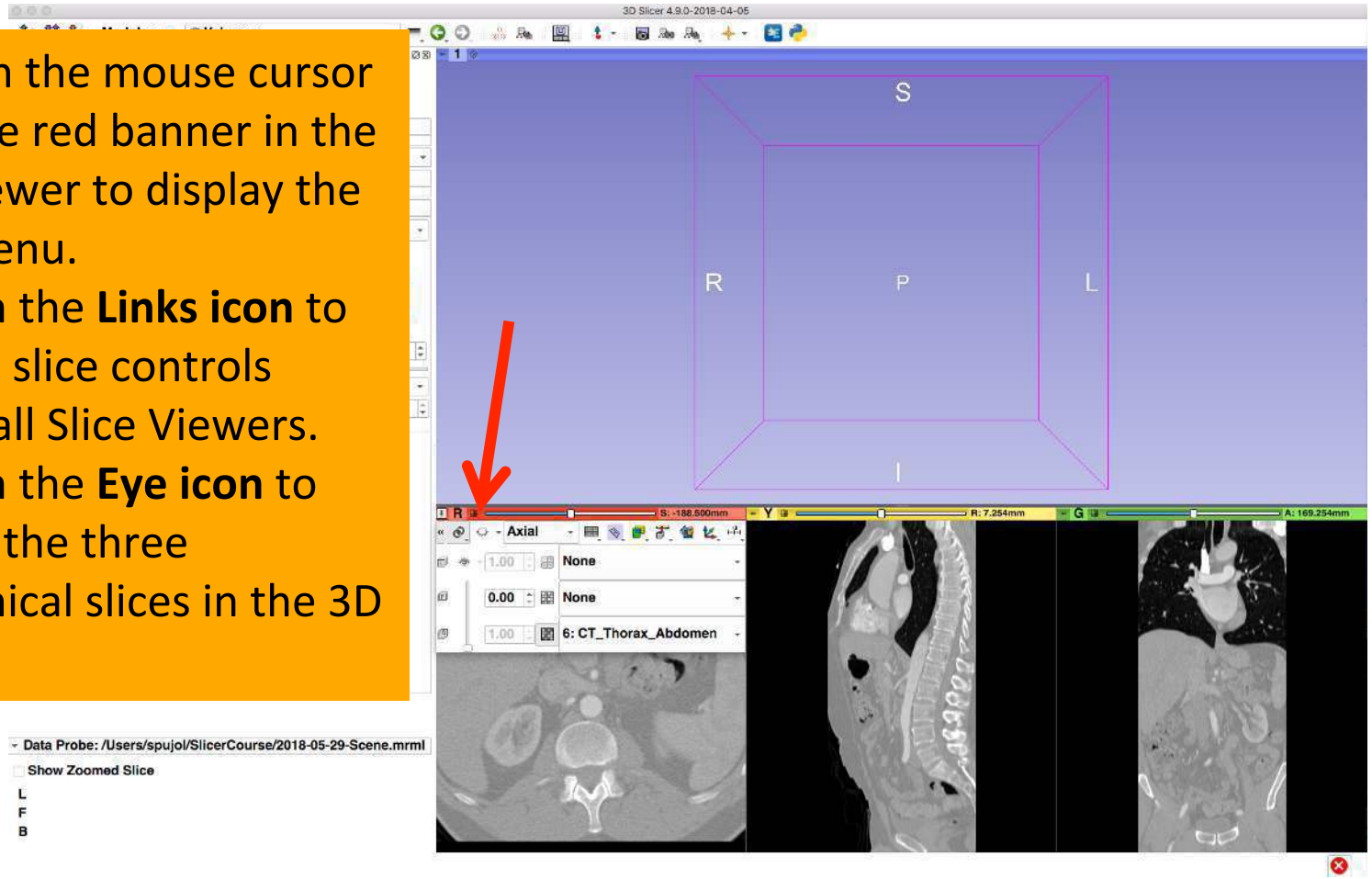
Select the Active Volume **6:CT_Thorax_Abdomen**

Click on the Window Level Preset **CT-abdomen** to automatically adjust the window/level display of the CT scan

© Sonia Pujol, Ph.D. All Rights Reserved

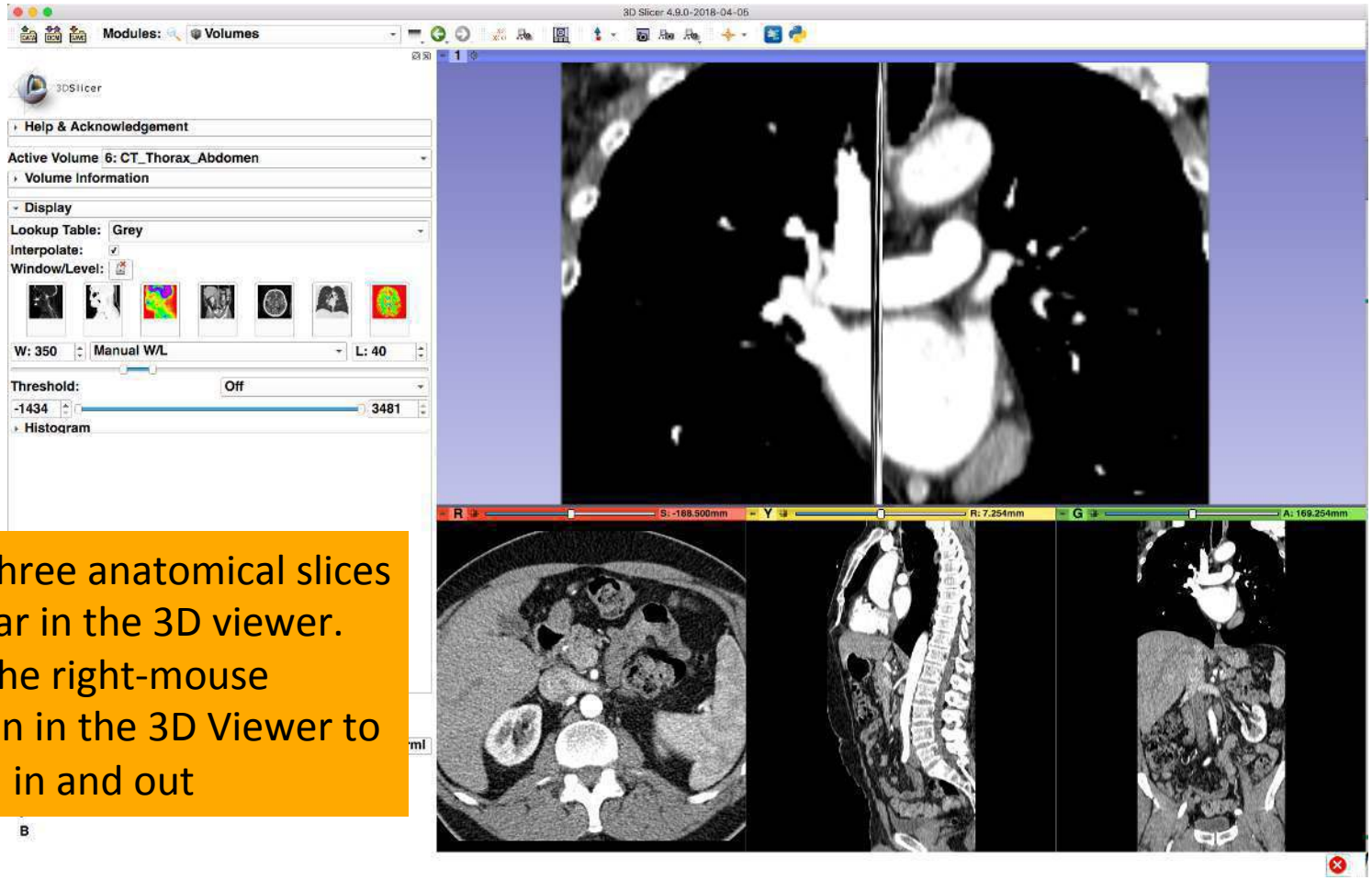
Loading a DICOM volume

Position the mouse cursor over the red banner in the Red Viewer to display the slice menu.
Click on the **Links icon** to link the slice controls across all Slice Viewers.
Click on the **Eye icon** to display the three anatomical slices in the 3D viewer



© Sonia Pujol, Ph.D. All Rights Reserved

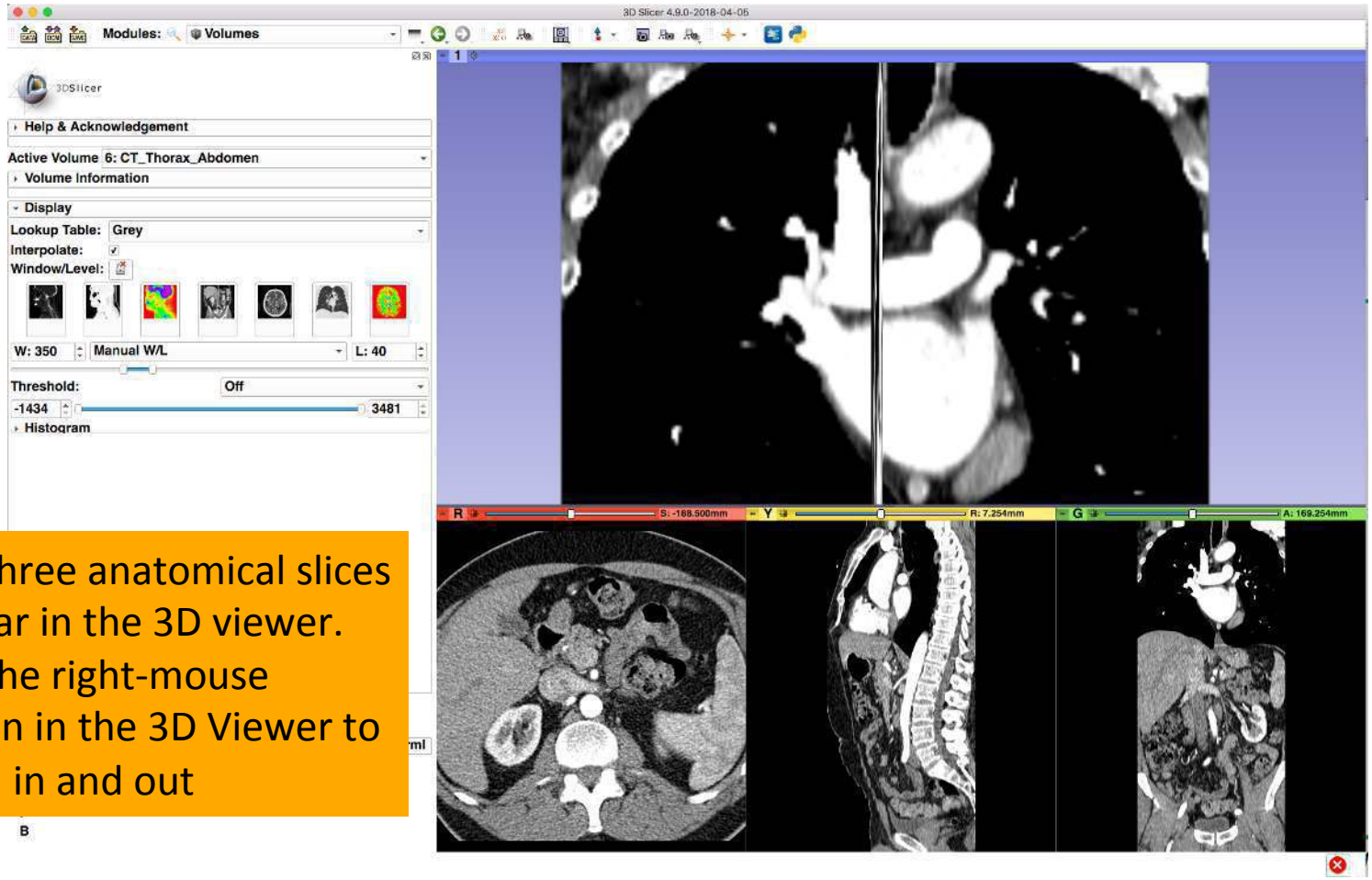
Loading a DICOM volume



The three anatomical slices appear in the 3D viewer. Use the right-mouse button in the 3D Viewer to zoom in and out

© Sonia Pujol, Ph.D. All Rights Reserved

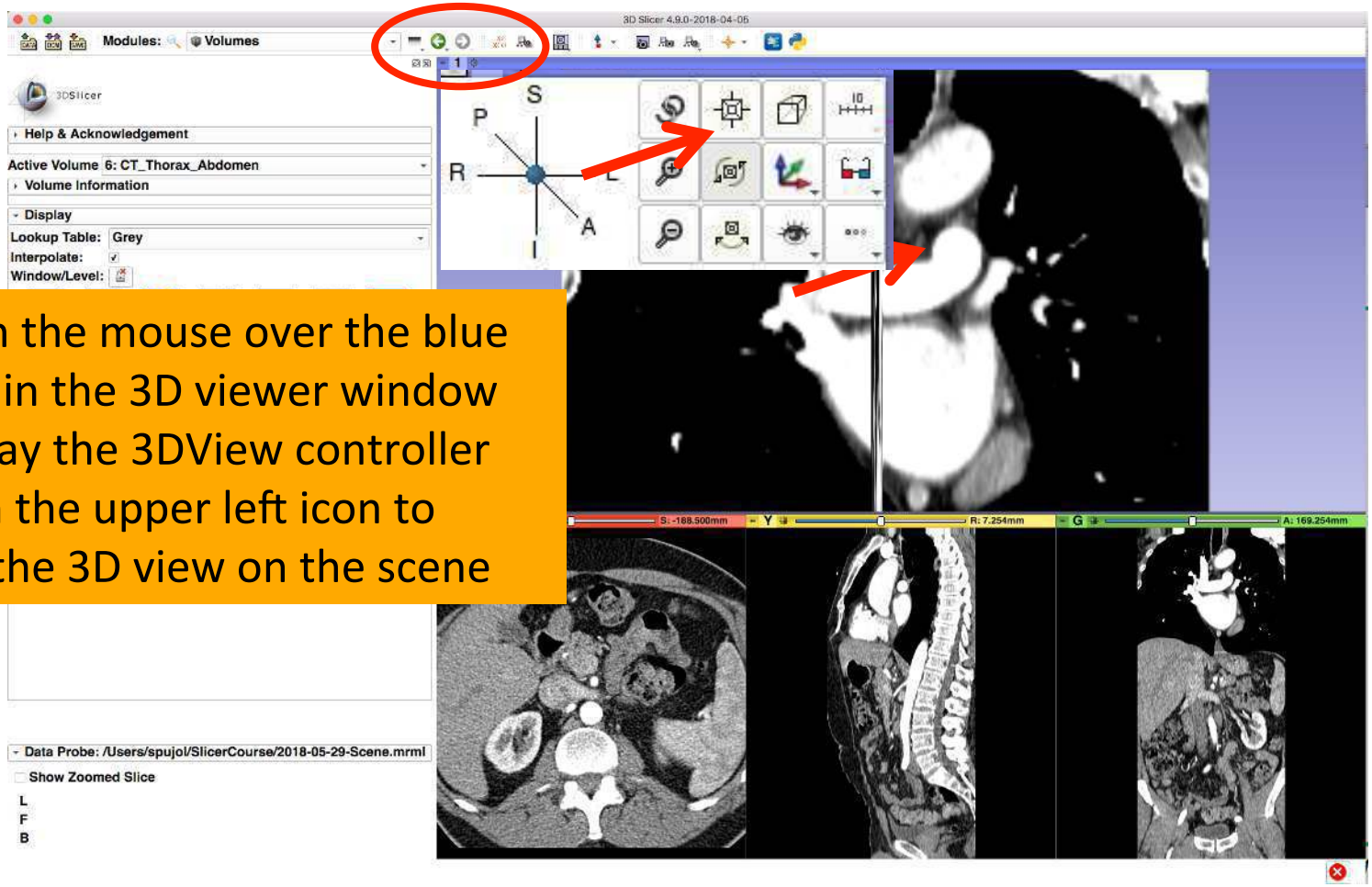
Loading a DICOM volume



The three anatomical slices appear in the 3D viewer. Use the right-mouse button in the 3D Viewer to zoom in and out

© Sonia Pujol, Ph.D. All Rights Reserved

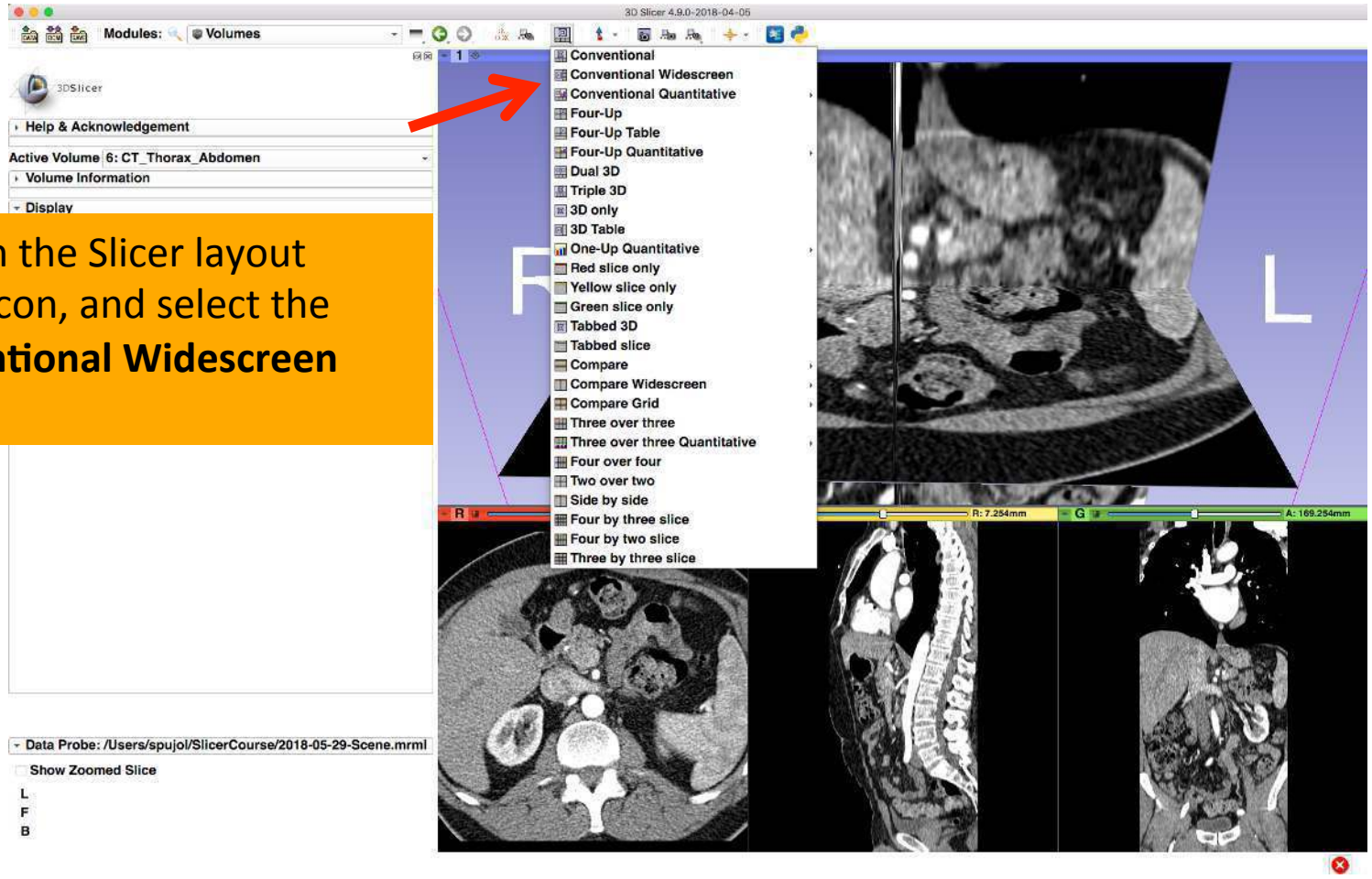
Loading a DICOM volume



Position the mouse over the blue banner in the 3D viewer window to display the 3DView controller
Click on the upper left icon to center the 3D view on the scene

© Sonia Pujol, Ph.D. All Rights Reserved

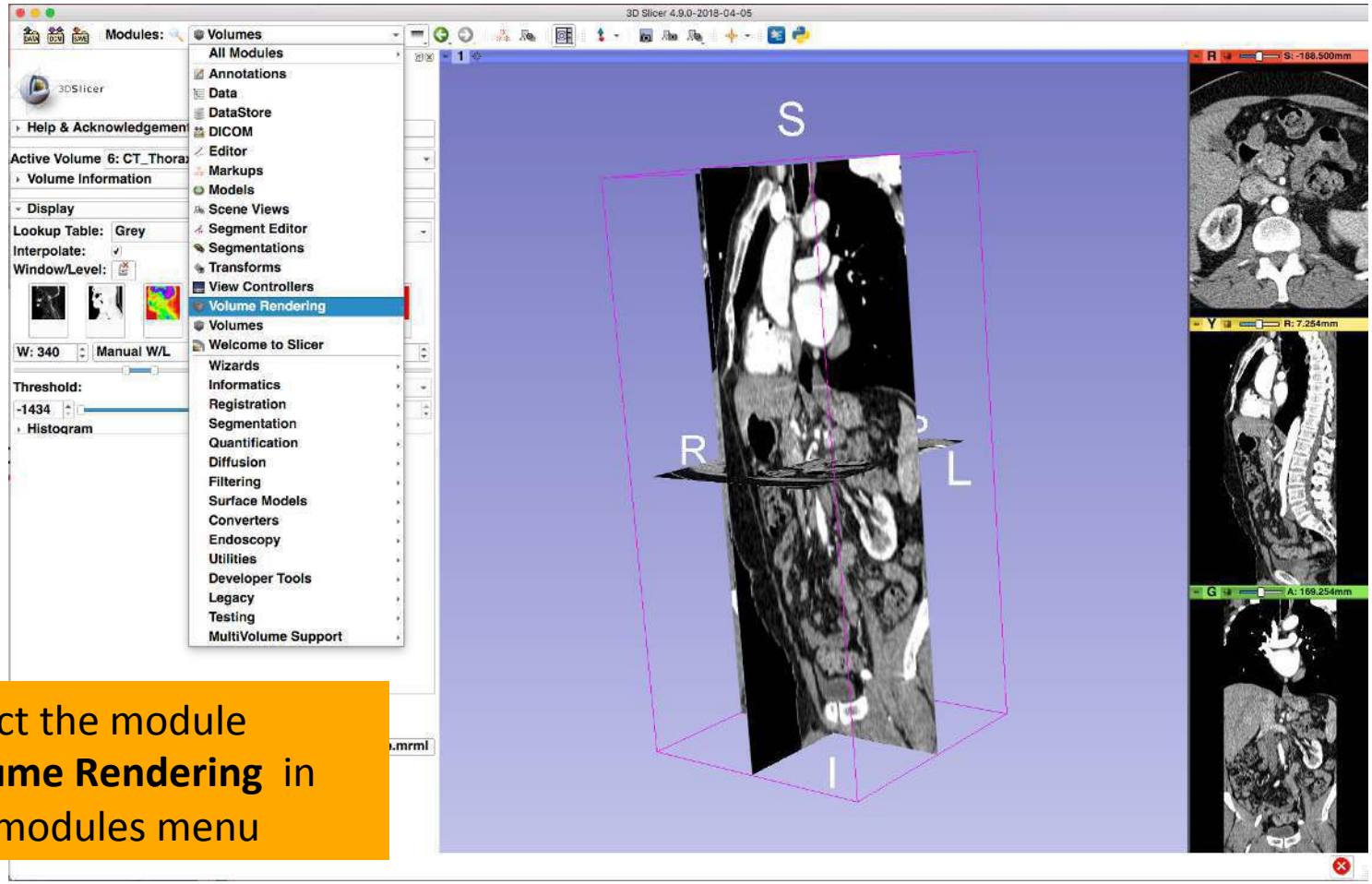
Loading a DICOM volume



Click on the Slicer layout menu icon, and select the **Conventional Widescreen** layout

© Sonia Pujol, Ph.D. All Rights Reserved

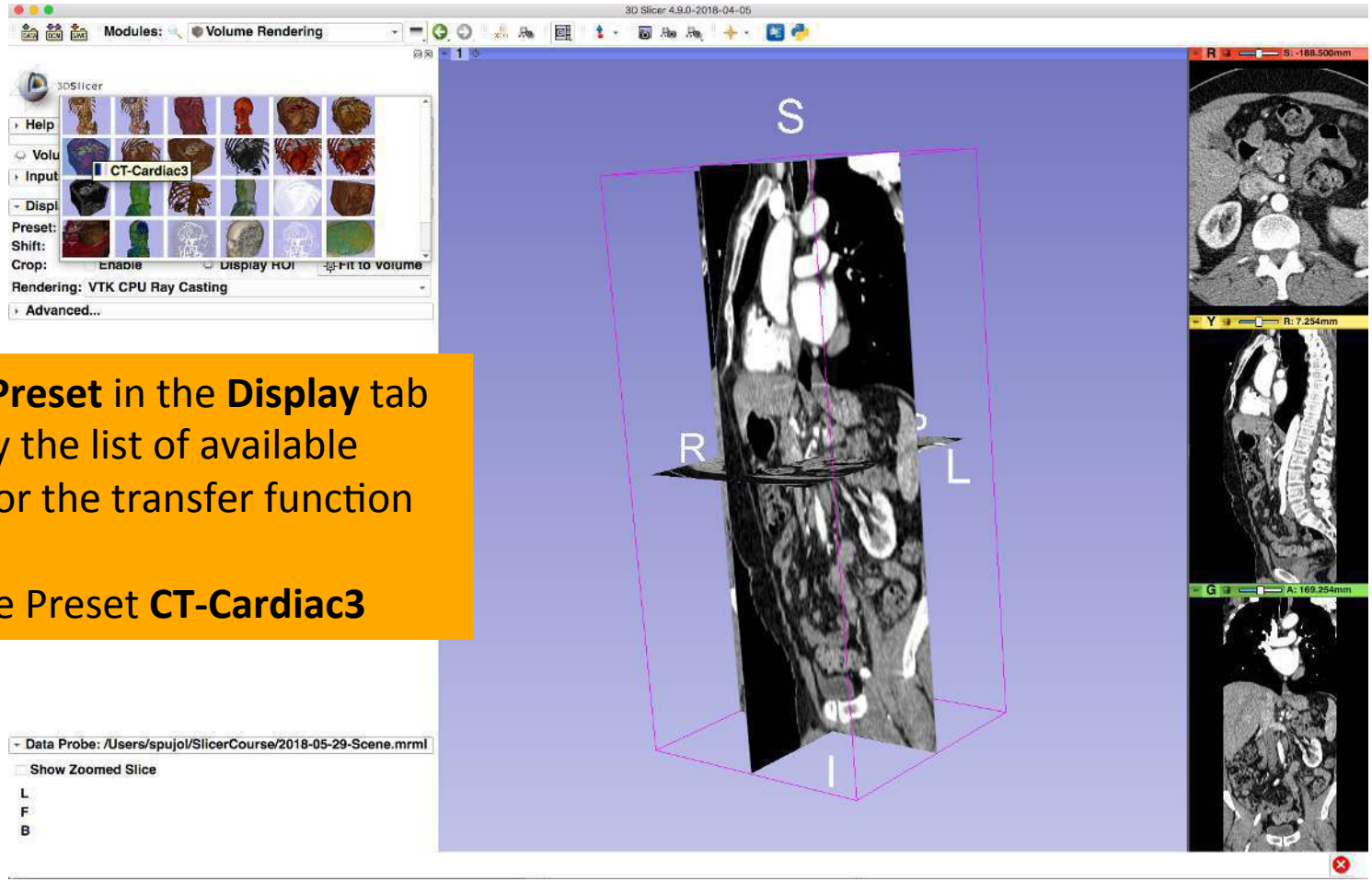
Volume Rendering



Select the module **Volume Rendering** in the modules menu

© Sonia Pujol, Ph.D. All Rights Reserved

Volume Rendering

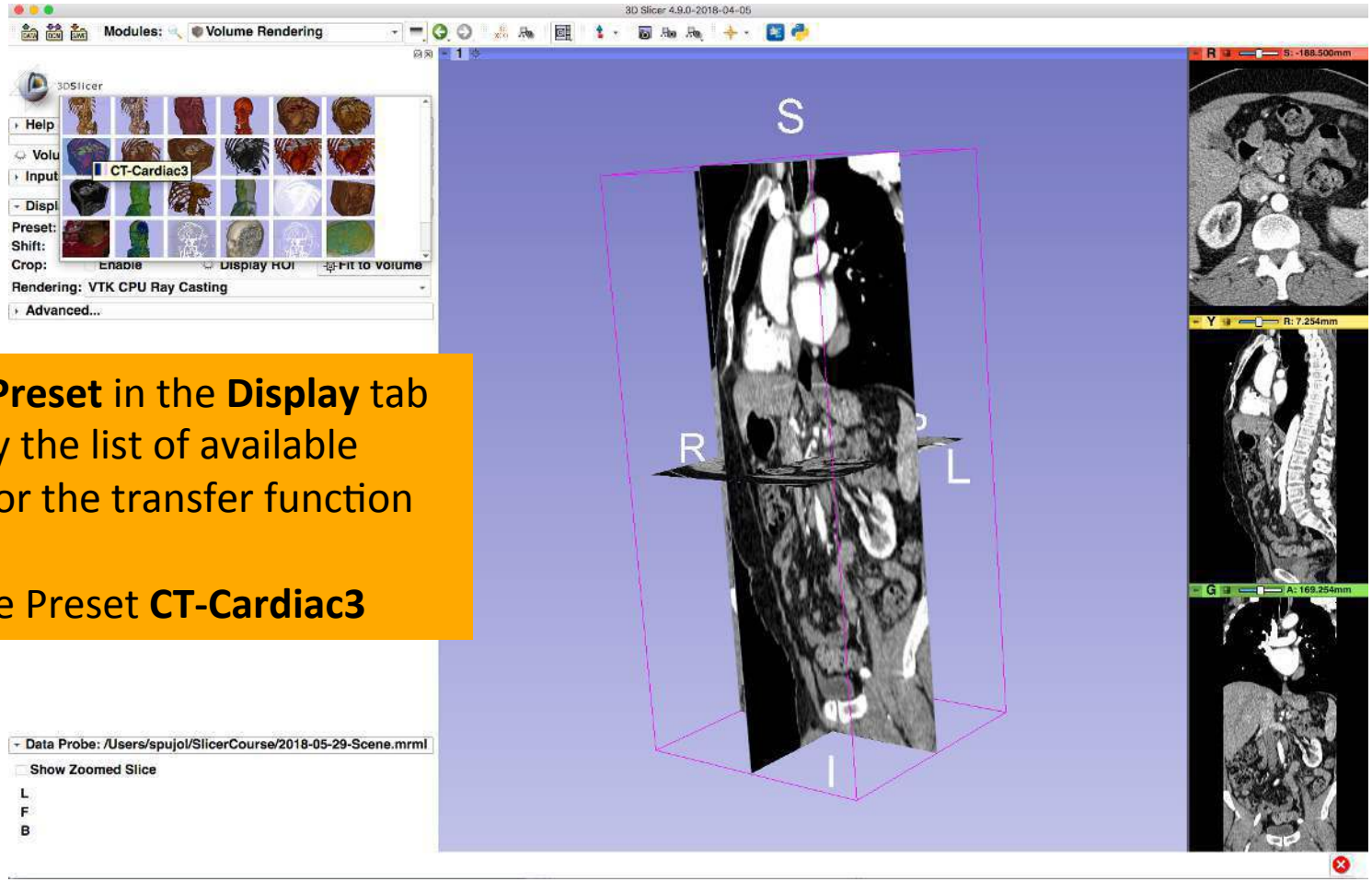


Click on **Preset** in the **Display** tab to display the list of available presets for the transfer function

Select the Preset **CT-Cardiac3**

© Sonia Pujol, Ph.D. All Rights Reserved

Volume Rendering

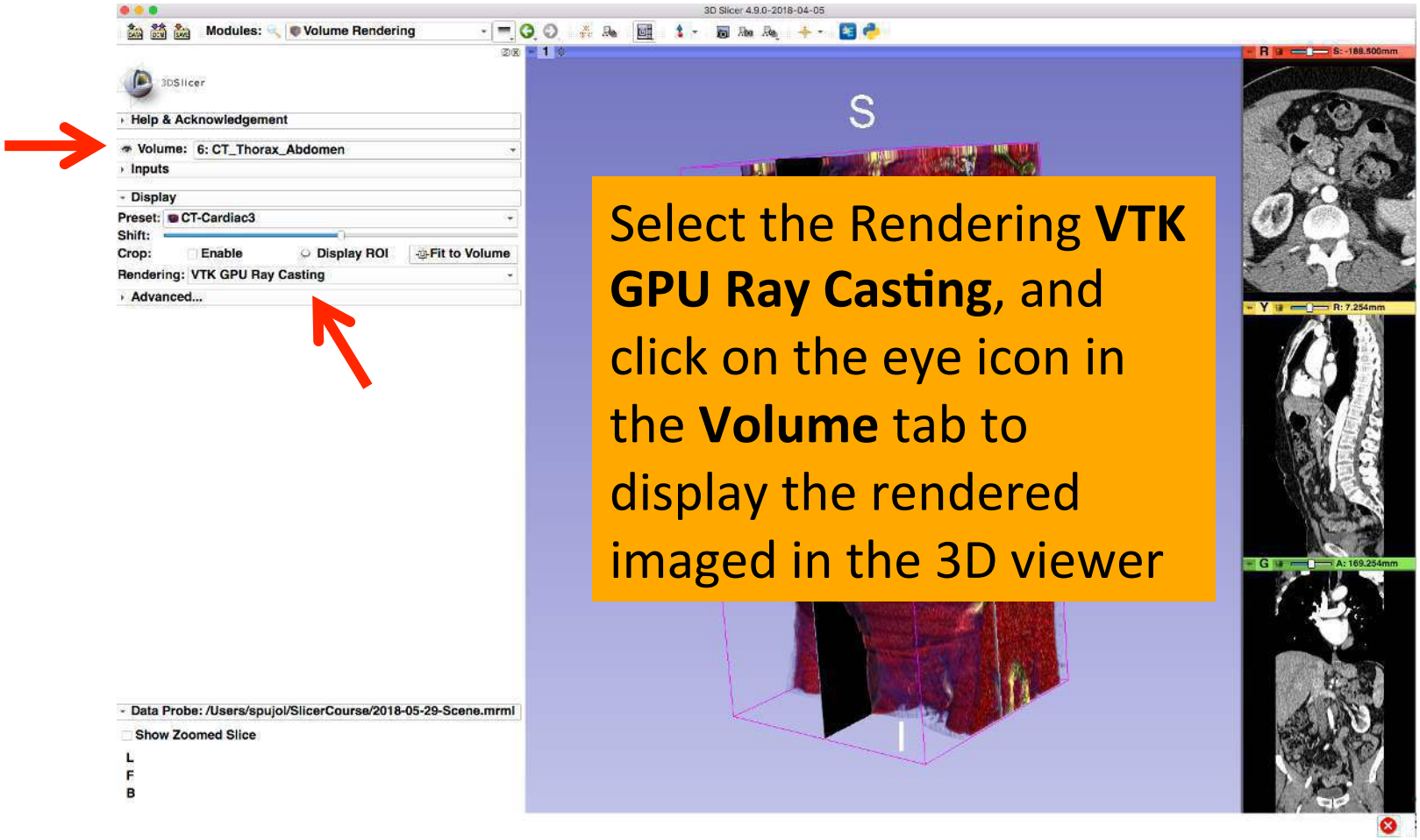


Click on **Preset** in the **Display** tab to display the list of available presets for the transfer function

Select the Preset **CT-Cardiac3**

© Sonia Pujol, Ph.D. All Rights Reserved

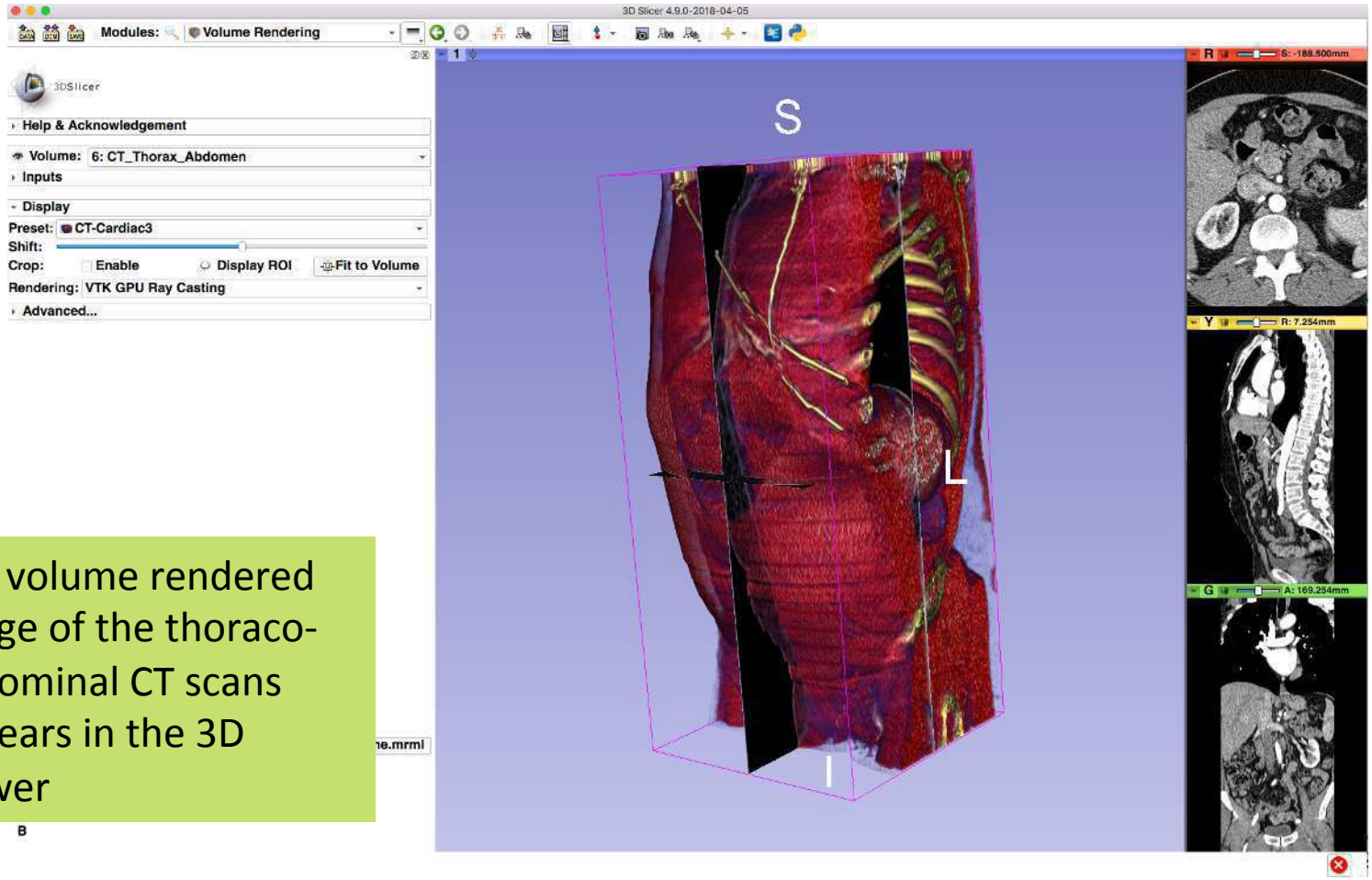
Volume Rendering



Select the Rendering **VTK GPU Ray Casting**, and click on the eye icon in the **Volume** tab to display the rendered imaged in the 3D viewer

© Sonia Pujol, Ph.D. All Rights Reserved

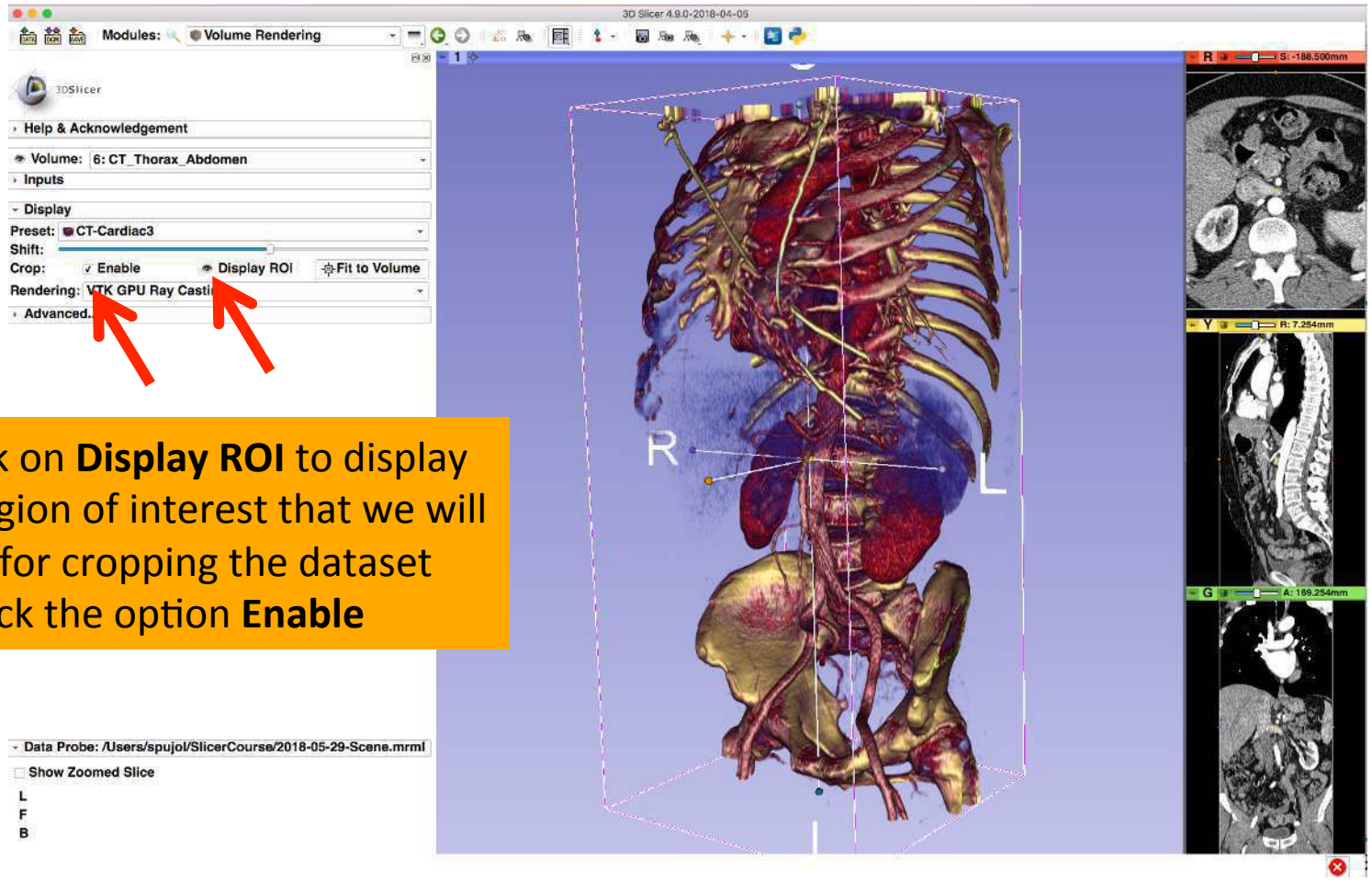
Volume Rendering



The volume rendered image of the thoraco-abdominal CT scans appears in the 3D viewer

© Sonia Pujol, Ph.D. All Rights Reserved

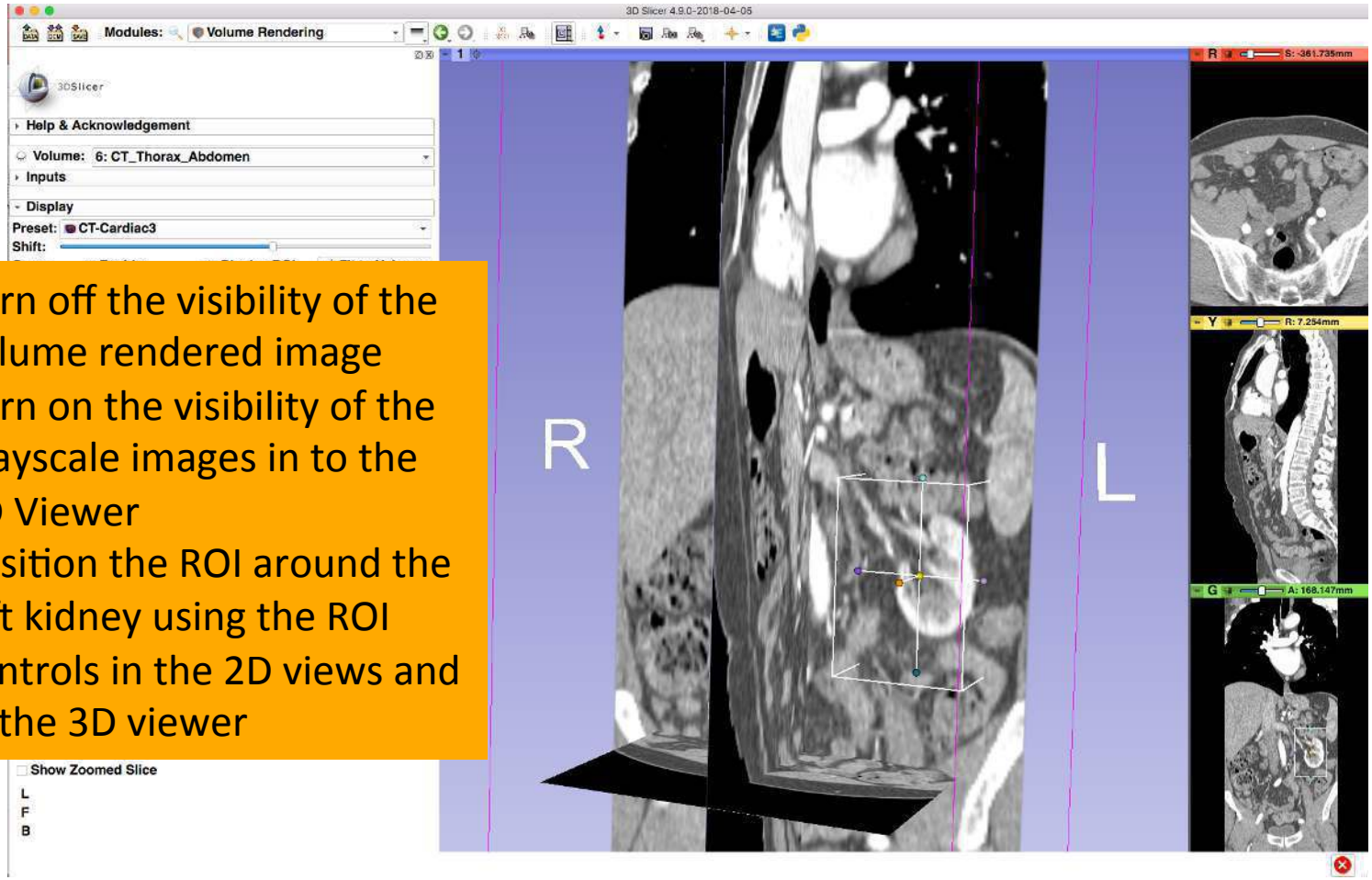
Volume Rendering



Click on **Display ROI** to display a region of interest that we will use for cropping the dataset
Check the option **Enable**

© Sonia Pujol, Ph.D. All Rights Reserved

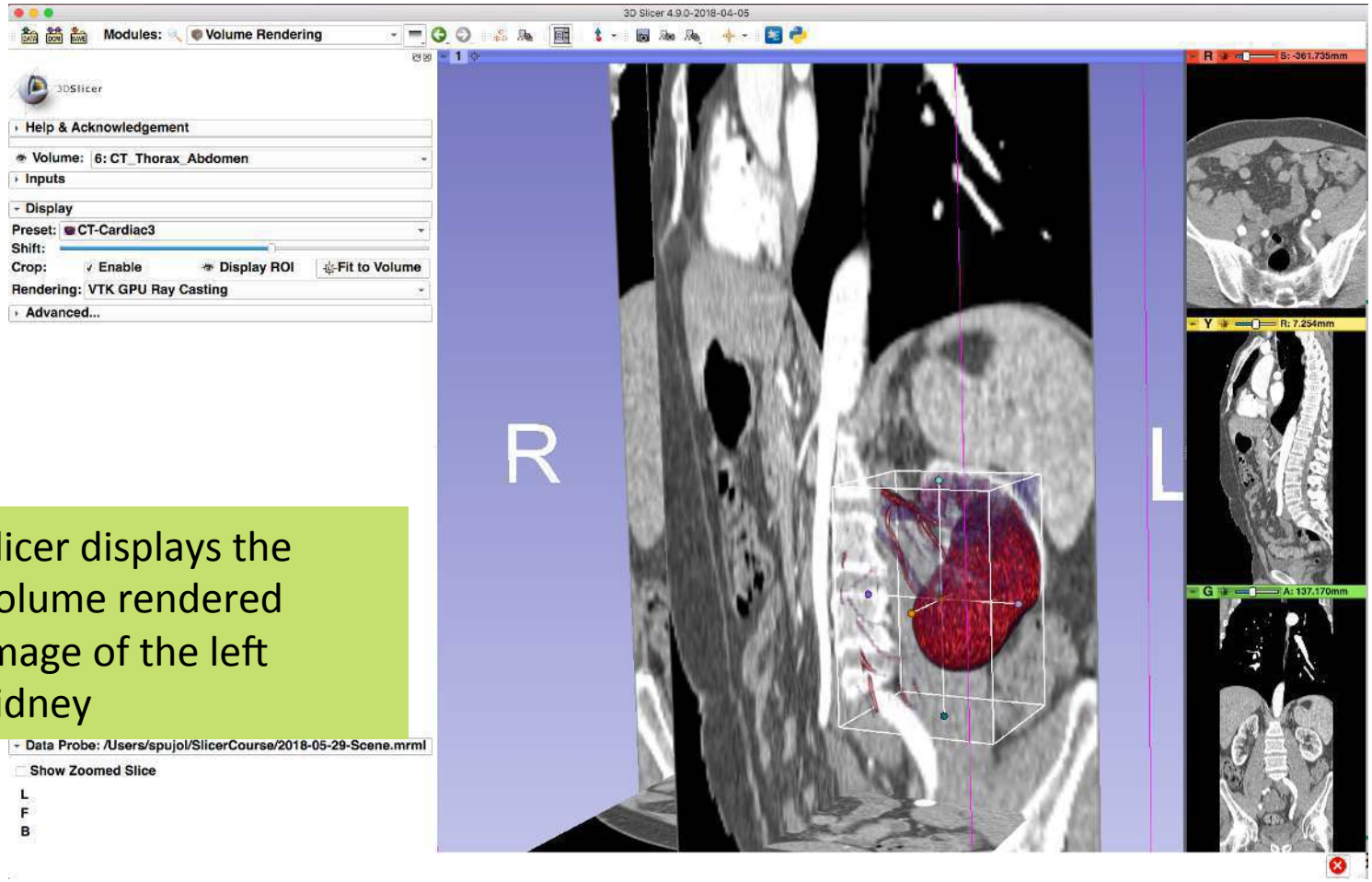
Volume Rendering



Turn off the visibility of the volume rendered image
Turn on the visibility of the grayscale images in to the 3D Viewer
Position the ROI around the left kidney using the ROI controls in the 2D views and in the 3D viewer

© Sonia Pujol, Ph.D. All Rights Reserved

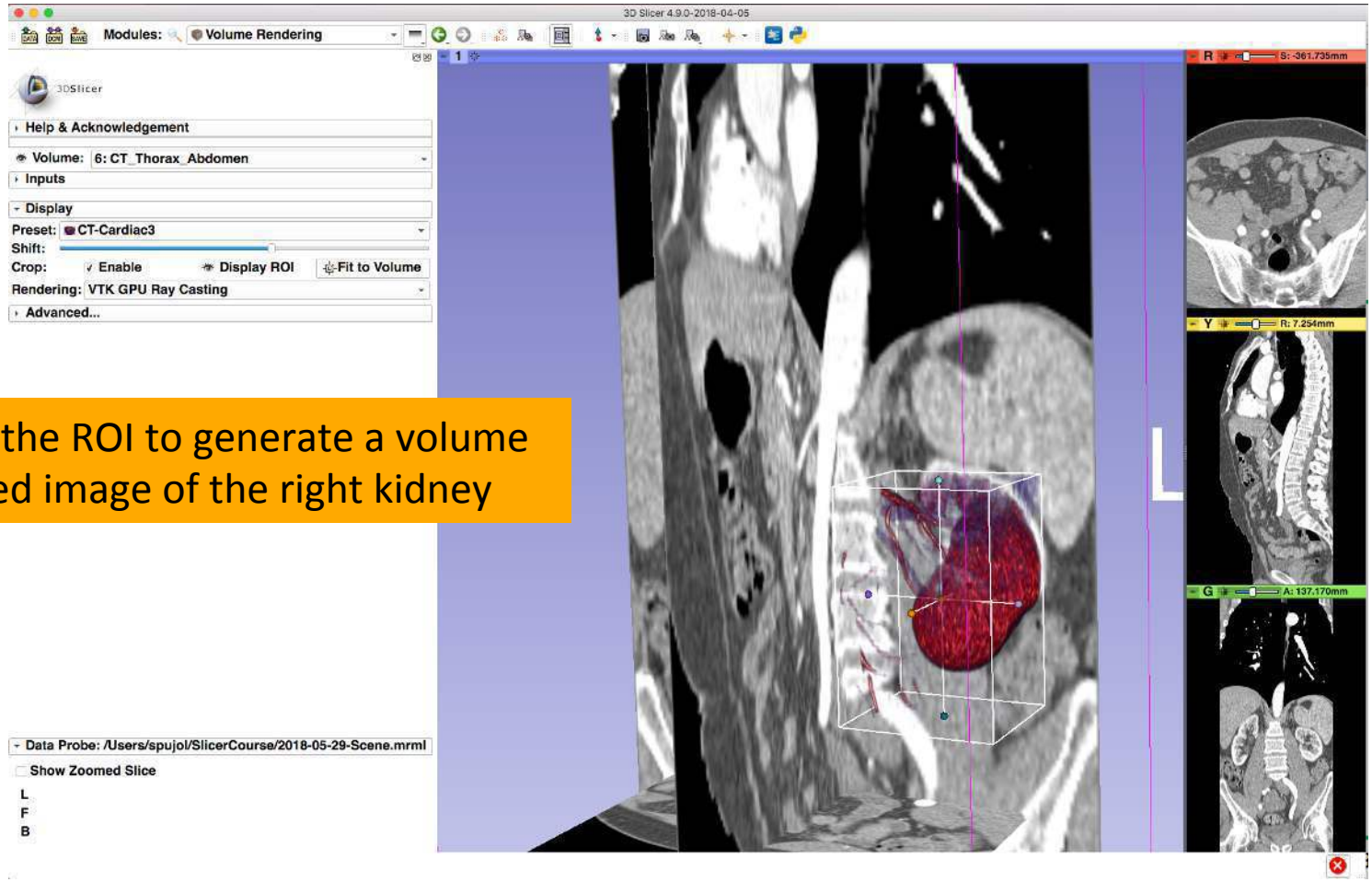
Volume Rendering



Slicer displays the volume rendered image of the left kidney

© Sonia Pujol, Ph.D. All Rights Reserved

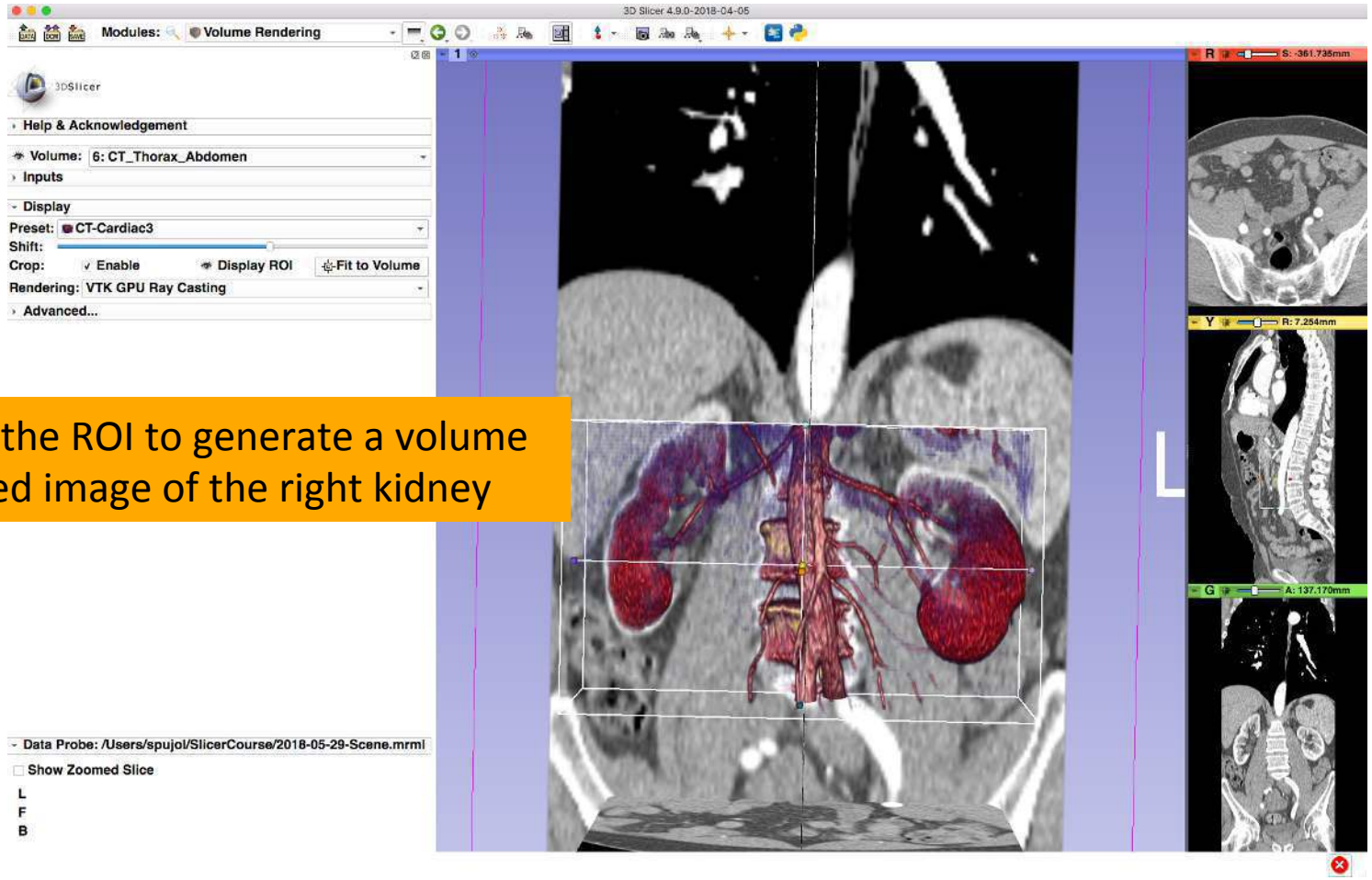
Volume Rendering



Extend the ROI to generate a volume rendered image of the right kidney

© Sonia Pujol, Ph.D. All Rights Reserved

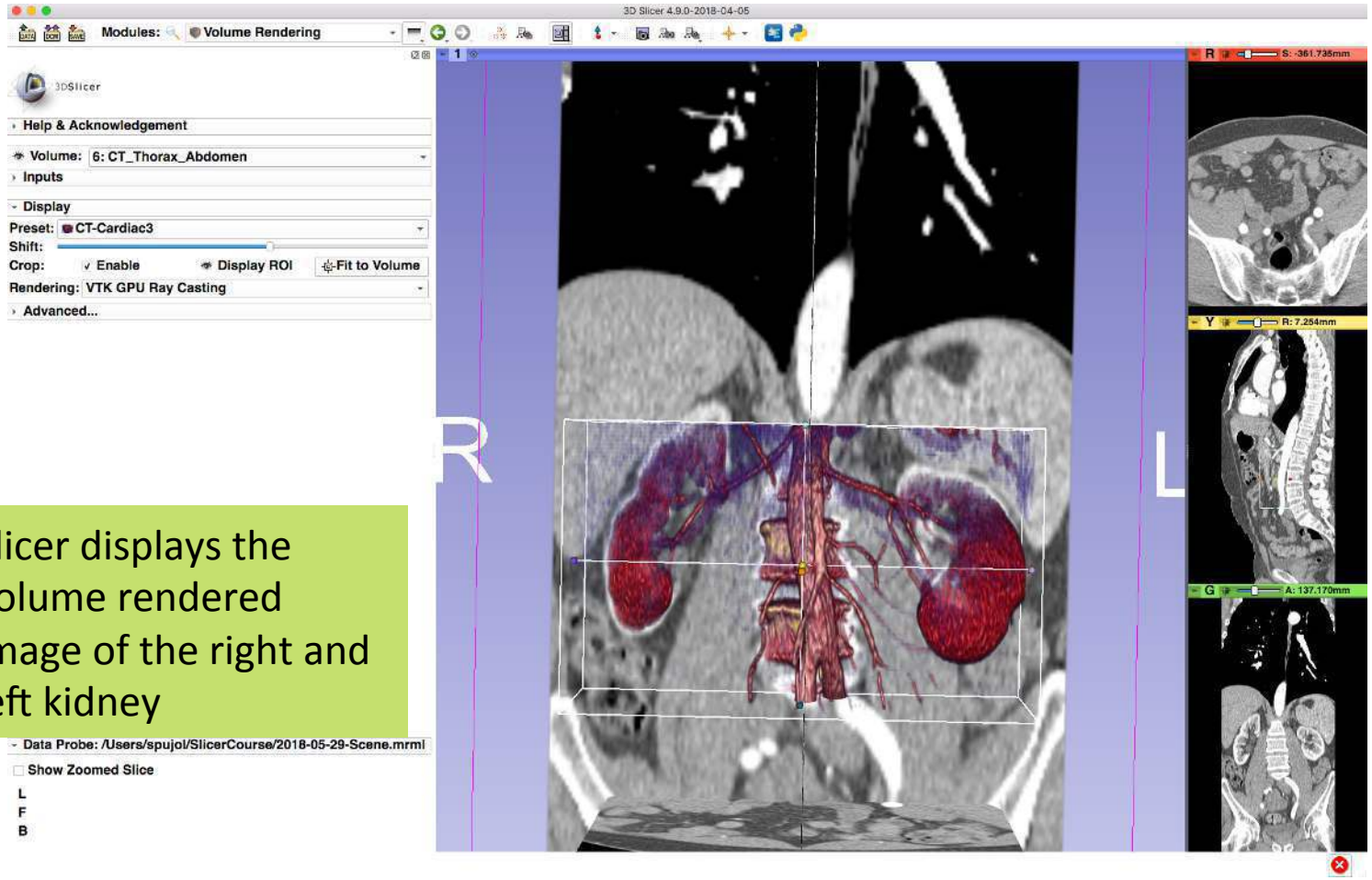
Volume Rendering



Extend the ROI to generate a volume rendered image of the right kidney

© Sonia Pujol, Ph.D. All Rights Reserved

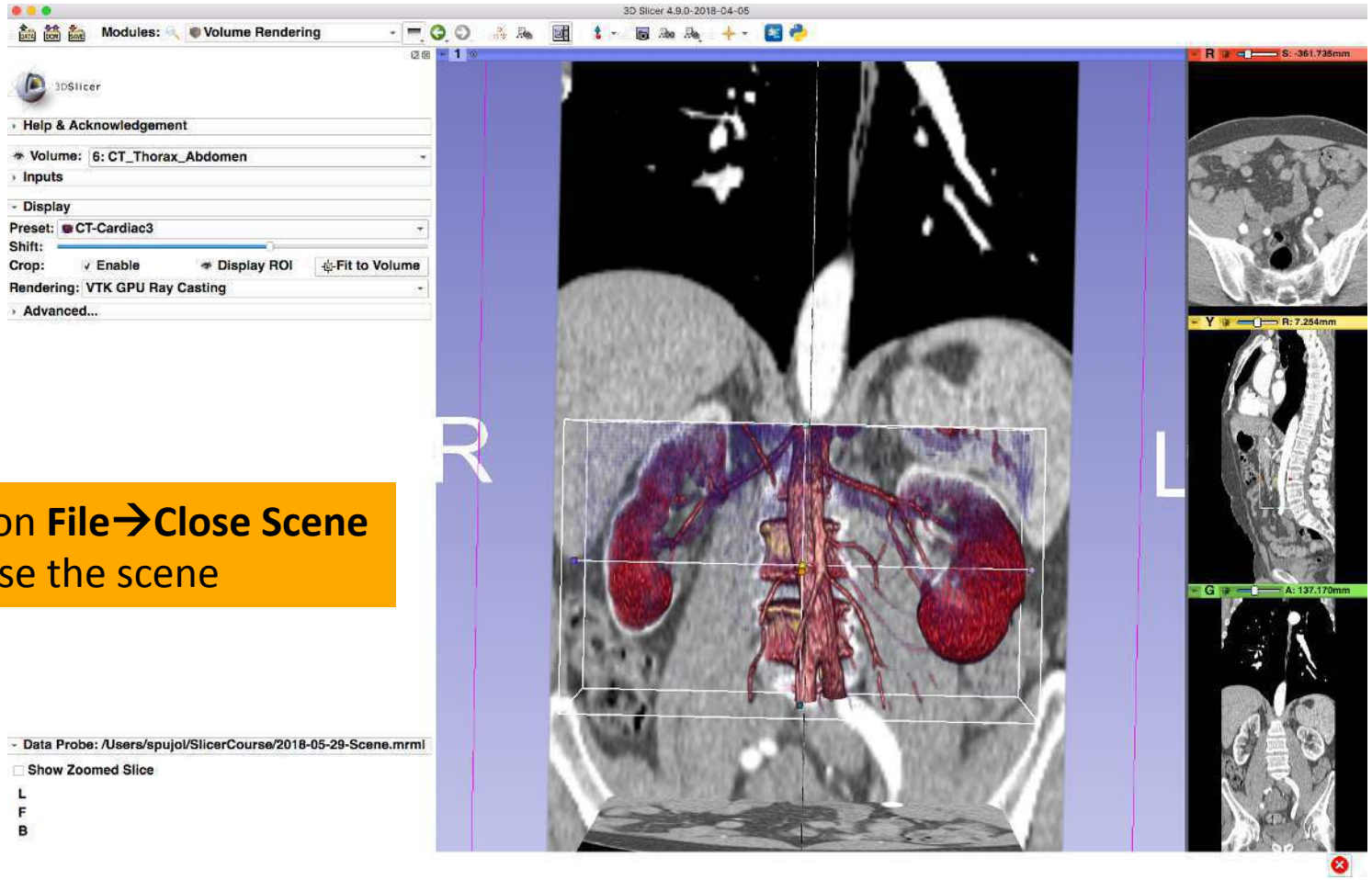
Volume Rendering



Slicer displays the volume rendered image of the right and left kidney

© Sonia Pujol, Ph.D. All Rights Reserved

Volume Rendering



Click on **File→Close Scene** to close the scene

© Sonia Pujol, Ph.D. All Rights Reserved

Acknowledgments

National Alliance for Medical Image Computing (NA-MIC)

NIH U54EB005149

Neuroimage Analysis Center (NAC)

NIH P41EB015902

© Sonia Pujol, Ph.D. All Rights Reserved